

Forest fuel structure and loading along a gradient of gray-phase mountain pine beetle severity in Jasper National Park, Alberta, Canada

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Abstract

Wildland fire fuels within the gray-phase of mountain pine beetle (*Dendroctonus ponderosae* Hopkins) outbreak (4–10 years post mortality) are treated as homogenous despite potential variability. To assess if proportion of mortality due to mountain pine beetle in gray-phase stands significantly affects fuel structure, composition, and loading, detailed fuel measurements were collected across a range of severity in Jasper National Park, Alberta, Canada, by coincidence 1–3 years before a major wildfire. These were categorized based on outbreak severity thresholds of Light (<35%), Moderate (36%–65%), and Severe (>65%) beetle-induced pine mortality defined as the proportion of killed trees within a stand. The canopy fuels of both mature and small tree cohorts were significantly reduced with increasing outbreak severity. Severely affected stands showed a 49% reduction in mature tree canopy bulk density from original loading but increased herbaceous vegetation. Moderately affected stands had higher coarse woody debris loading (mean = 3.21 kg m⁻²; SE = 0.49 kg m⁻²). These severity ratings can be used to quantify canopy fuel loss and signal potential shifts to other surface fuel types, barring fire. This is one of the first studies to showcase variability within gray-phase outbreak fuels, despite their prevalence in western Canada, and discuss the potential influences on fire behaviour.

Key words: wildfire fuels, insect outbreak severity, forest disturbance, regeneration, fire behaviour

Preface

On July 22, 2024, lightning ignited a wildfire in Jasper National Park 28 km south of the community of Jasper in western Alberta, Canada. Two days later, the fire reached the town and ultimately destroyed 358 residences and other structures.¹ By coincidence, the present study involved measuring the fuel structure of the conifer forests of the Athabasca Valley, with the majority of fuel plots contained within the fire perimeter. While the present analysis was completed beforehand, the fire and damage greatly heightened the immediate importance of this fuel survey work. Fuel structure and composition are a critical element of the fire environment (Keane 2015), and the findings from this study are currently being examined by other researchers studying the Jasper fire event.

Introduction

The recent mountain pine beetle (*Dendroctonus ponderosae*; MPB) outbreak affected over 18 million hectares in western

Canada (Natural Resources Canada 2022) and has led to an abundance of fuel complexes that differ from natural, undisturbed stands. These altered fuels may result in wildland fire behaviour that is not readily predicted by traditional models (Dhar et al. 2016). The legacies of MPB outbreaks are typically classified by “time since beetle” phases (i.e., green, red, gray, and old-phase) (Schoennagel et al. 2012). Variation in fuel moisture, chemistry, and within-stand biomass distribution across these phases have been well documented (e.g., Klutsch et al. 2009; Jenkins et al. 2014). However, within the gray-phase (4–10 years post mortality), the level of MPB outbreak severity (as the proportion of killed trees out of total tree stems), potentially has highly variable effects on stand structure and the spatial distribution of dry fuels across the landscape. Gray-phase stands are characterized by near total foliage loss, deterioration and surface fuel accumulation of fine fuels such as small twigs and bark, while dead trees (i.e., snags) remain standing (Klutsch et al. 2009; Schoennagel et al. 2012). These changes to the condition and distribution of biomass within a stand can increase ignition potential and surface fire intensity, and create difficult-to-action scenarios

¹ <https://jasper-alberta.ca/p/damagemap>

for wildland fire fighters when ignitions do occur (Page et al. 2013; Moriarty et al. 2019).

MPB outbreaks affect fuels in complex ways that are difficult to accurately represent within traditional fire behaviour models (Schoennagel et al. 2012; Jenkins et al. 2014). This difficulty has led to a lack of consensus on gray-phase MPB fuel effect on fire behaviour, with equal responses of amplified, neutral, and dampened fire responses found in simulated and observational studies reviewed by Romauldi et al. (2023). Many studies (e.g., Jenkins et al. 2008; Hicke et al. 2012) have suggested that the hazard of crown fire potential in gray-phase stands could be reduced due to the loss of canopy fuels. However, this effect might be counteracted by an increase in rate of spread in surface fire due to increased solar radiation and within stand wind speeds (Jenkins et al. 2008; Whitehead et al. 2008), higher amounts of cured grass or herbaceous material in the spring and fall (Page and Jenkins 2007), and surface fire intensity due to coarse woody debris loading (Schoennagel et al. 2012). Instances of high severity crown fire have also been found in gray-phase outbreak despite low canopy fuel load; highlighting the variation in fire behaviour that can emerge in affected stands (Agne et al. 2016; Perrakis et al. 2014). Talucci and Krawchuck 2019 found that branch structure consumption and char depth increased with outbreak severity in extreme fire weather conditions, indicating higher fuel availability in gray-phase stands. Sloughing bark and compromised structural integrity can also increase spotting and firebrands. Moreover, wildland firefighters have reported unexpected, sustained crown fire and increased hazards due to the unpredictable nature of falling snags in gray-phase MPB stands (Moriarty et al. 2019).

The forests of Jasper National Park, on the eastern slopes of the Rocky Mountains in central Alberta, were heavily affected by the recent MPB outbreak. An estimated 66 000 hectares (approximately 55% of the total majority lodgepole pine stands within the park) experienced mortality from MPB by the end of the outbreak (British Columbia Forest Analysis and Inventory Branch 2024). The outbreak ended due to a MPB mortality event induced by prolonged periods of extreme cold weather (i.e., $<-30^{\circ}\text{C}$) in the winters of 2018 and 2019 (Parks Canada 2020, unpublished report). Most MPB-affected stands were in the gray-phase state as of 2021. In Jasper, a decrease in fire frequency over the last 100 years or more due to colonial fire suppression has led to a shift from a historical mosaic of conifer, aspen (*Populus tremuloides* Michx.), open Douglas-fir (*Pseudotsuga menziesii* Franco) stands, as well as open grasslands (Tande 1979), to denser and more homogeneous lodgepole pine- and spruce-dominated forests (Rhemtulla et al. 2002). Suppression-induced shifts in fire regimes in fire-regulated ecosystems have been reported in this and other montane ecoregions of the Canadian Rockies (Rhemtulla et al. 2002; Chavardès and Daniels 2016). The absence of fire in these areas has increased the prevalence of large areas of closed canopy pine forest containing shade-tolerant understory species (e.g., spruce and fir) that form ladder fuels (Chavardès and Daniels 2016). These large areas of homogenous and mature lodgepole pine forest created an ideal scenario for high-severity MPB outbreaks (Raffa et al. 2008). The legacies of this outbreak (i.e., gray-phase for-

est), coupled with a warming climate, led to significant increased concern over elevated wildfire risk throughout the park.

In this study, we investigate how wildland fuels are affected by MPB outbreaks in the montane forests in Jasper National Park. Specifically, we investigate how gray-phase post-MPB fuels differ from other conifer fuel types and how MPB outbreak severity, stand density, and forest regeneration affect gray-phase fuel structure. Establishing specific disturbance baseline fuels in altered fuel complexes is necessary to evaluate fire hazard potential as well as ecosystem change at large. Improved understanding of these complex spatial and temporal interactions among insect outbreaks, fuels, and wildland fire potential is of increasing importance as climatic warming increases wildland fire potential across western Canada.

Materials and methods

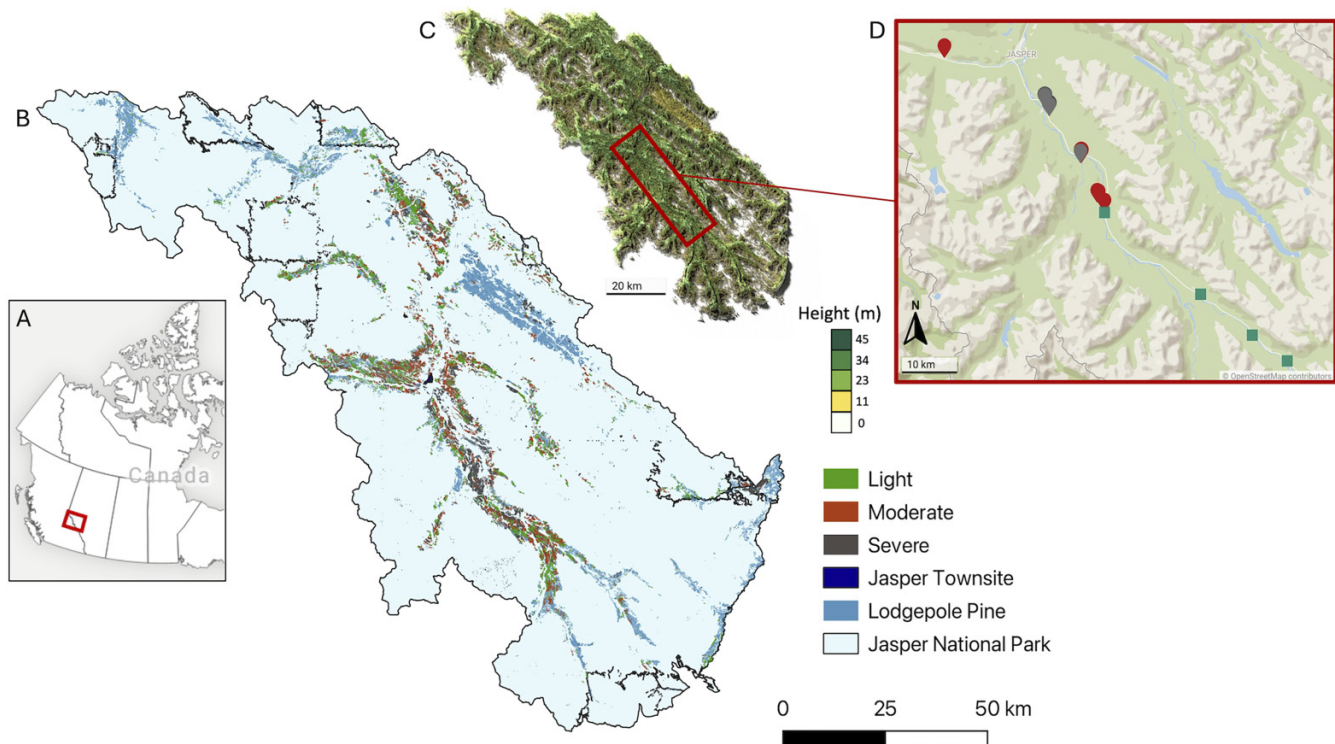
Study area

Jasper National Park (52.8 N, -117.9° W) is located on the eastern slopes of the Canadian Rocky Mountains along the western border of central Alberta, Canada (Fig. 1). This region is within Treaty 6 and 8 lands, as well as the traditional territory of the Anishinabe, Aseniwuche Winewak, Cree, Nêhiyawak, Stoney Nakoda, Secwépemc, Dene-zaa, Mountain Métis and Métis peoples for thousands of years. Jasper encompasses an area of 11 000 km² and includes montane, subalpine, and alpine ecotypes. The montane region is dominated by lodgepole pine with areas of Douglas-fir or hybrid spruce (*Picea glauca* Voss $\times$ *Picea engelmannii* Parry) but also includes white spruce (*P. glauca*) and aspen interspersed with grassland (Chavardès and Daniels 2016). Jasper is in the Montane-Cordillera ecozone of Canada, with a subarctic climate, and is in the rain-shadow of the Canadian Rockies. Climate normals from 1981 to 2010 show monthly mean temperatures range from -6.9°C to 15.3°C with 599 mm of mean cumulative annual precipitation (Jasper Warden Station, Environment Canada 2022). The Montane-Cordillera region experiences the highest density per hectare of combined lightning and human-caused wildfires in the country, which presents a continuous threat to inhabited residential areas (Coogan et al. 2020).

Study design

Potential locations for forest mensuration-based fuel plots were initially identified using high resolution satellite images (Google Earth Pro 2021) followed by physical reconnaissance to ensure the range of MPB severity was captured. Sites selected were all lodgepole pine-dominated and with little to no topographic relief, located in the montane Athabasca Valley. Plots were categorized into outbreak severity threshold bins (e.g., Wulder et al. 2009) scaled to Light (0%–35% mortality), Moderate (36%–65% mortality), and Severe (66%–100% mortality) based on the proportion of dead trees within each plot (Fig. 2). These severity classes allowed for spatial comparison to the vegetation resource inventory vector GIS layer provided by Jasper National Park (British Columbia Forest Analysis and Inventory Branch 2024) which categorized MPB

Fig. 1. Map of study area (A) Jasper National Park's location within Alberta, Canada. (B) Spatial extent of each mountain pine beetle severity class across the park. (C) Jasper National Park coloured with canopy height estimated from 10 m resolution tree height distribution from 2020 Sentinel-2 imagery fused with GEDI LiDAR vertical distribution data (Lang et al. 2022). White areas indicate non-vegetated areas. Inset (D) shows locations of 18 Mountain Pine Beetle fuel mensuration plot pairs including Light outbreak (green squares; $n = 5$), Moderate outbreak (red markers; $n = 8$), and Severe outbreak (gray markers; $n = 5$). Figs was created using QGIS version 3.34, R version 4.3.1, and Datawrapper version 3 using data assembled from the following sources: **British Columbia Forest Analysis and Inventory Branch 2024**, OpenCanada (opencanada.ca), Google Earth Engine Community Catalog (gee-community-catalog.org). Base map from OSM (openstreetmap.org).



disturbance by “% code”. Light severity class corresponded with codes 1–3, Moderate with codes 4–6, and Severe with 7–9. All measured plots had some pine mortality from MPB, therefore the “Light” severity class is assumed as a reference as these stands were relatively undisturbed compared to the Moderate and Severe classes. The % mortality thresholds used here were adjusted from those used in **Wulder et al. (2009)**, as they classed severity levels as 11%–30% mortality = “moderate”, 31%–50% = “severe”, and >50% = “very severe”. **Wulder et al. (2009)** predicted that cumulative disturbance over time could require an increase of severity rating threshold values to better reflect the level of mortality after 5–10 years of continued infestation, which has since been seen elsewhere in western Canada. The higher mortality thresholds used in this study thus provide a relevant scale for any area experiencing prolonged MPB outbreak.

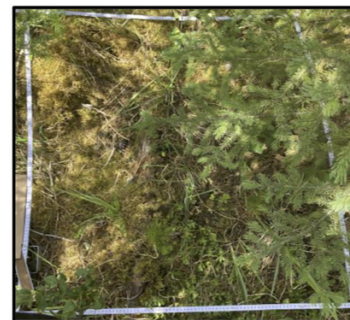
Field data collection

To quantify fuel variability across forest stands, detailed measurements were taken for each fuel stratum (ground, surface, understory, canopy) within forest mensuration plots, as outlined by **Boucher et al. (2023; Fig. 3)**. Twelve plots were measured in 2021, six plots in 2022, and two plots in 2023. Two plots from 2022 were excluded from analysis due to

the large proportion of deciduous trees (>50%), making fuel comparisons inappropriate between largely varying species compositions. A total of 18 valid plots were used for analysis. Each plot was centrally located using Global Navigation Satellite System (GNSS, Topcon, Inc. California, USA) receivers with sub-decimeter positional accuracy. Markers at 4 m and 11.28 m were staked out in each cardinal direction as distance from centre to delineate a 50 m² circular understory plot and 400 m² circular overstory plot, respectively. To measure surface fuel, 8 × 1 m² microplots were installed, where fuel bed depth and weight (of litter and fermented/humic layer), forest floor vegetation/detritus coverage by category and conifer seedling density were recorded. Downed woody debris was measured along 40 m transects separated by diameter and characterized by condition (sound or rotten) and position (elevated or on the ground) (**Boucher et al. 2023**). To measure understory fuel attributes, all saplings (DBH ≤ 3 cm), seedlings (height < 1.3 m), and dominant shrub and herbaceous species were measured within the 50 m² plot. To measure overstory fuel attributes, tree species, tree height, height to live crown, height to dead crown, DBH, and condition (percentage defoliated, live or dead, MPB status) were determined for all mature trees (DBH ≥ 9 cm) and small trees (DBH ≥ 3 cm) within the 400 m² plot.

Fig. 2. Examples of stand structure, canopy conditions, and surface vegetation composition for each mountain pine beetle severity class showing greater herbaceous material coverage in Severe plots. Photographs taken by author Tristan Skretting.

Light



Moderate



Severe

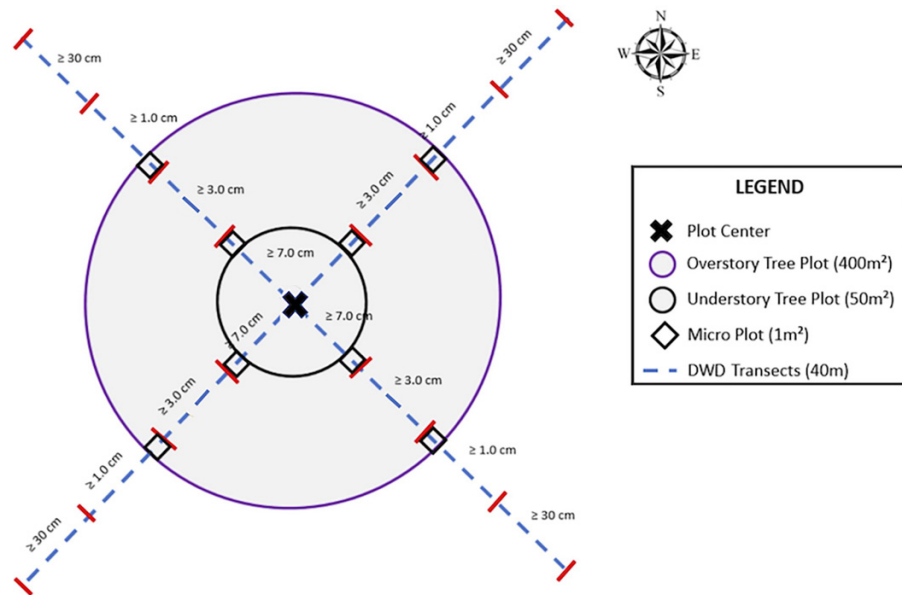


Data processing

Forest stand and fuel characteristics were calculated from measured plot data for each stand type. Stand density, basal area, quadratic mean diameter, tree height, height to the base of crown, species composition and MPB mortality proportions (assigned for all dead lodgepole pines within each plot) were averaged by outbreak severity rating. Crown fuel load and crown bulk density were calculated for all mature trees, small trees, and saplings using allometric equations for biomass from Lambert et al. (2005) based on DBH and tree height. These regression models provide estimates of biomass for wood (bole), bark, foliage, and branch components of trees for each species present (root-mean-square-error for foliage component 2.6–12.9 kg). Crown fuel load is the dry weight of the foliage and fine branches in the crown available for consumption in a fire, and canopy bulk density divides this weight by the crown length (Phelps et al. 2022). The definition of crown fuel load is not consistently applied among researchers and can include twigs, small branch wood up to 1 cm diameter or 3 cm diameter, and aerial lichens and mosses (Alexander and Cruz 2011), or needle foliage alone (Van Wagner 1977). Biomass equations separating

branch wood by diameter were not available in this case, so only conifer foliage and the smallest twigs were included in the initial calculation (Phelps et al. 2022). Canopy fuel load per unit area across the plot was calculated by tree status; standing trees in gray-phase (dead with no needles) were excluded from the canopy fuel load calculation, as complete loss of foliage was noted. To highlight the loss of canopy fuels due to MPB, foliage load was calculated for trees and saplings as if all were in live condition, and then condition was classified to show foliage reduction that has already taken place (gray trees) and soon to be lost (red trees) from the canopy. Though the stands were overall in gray phase, infestation is not homogenous as beetles affect trees in a staggered manner, and foliage drop rates can vary. Thus, trees that were killed by MPB most recently before measurement still retained some foliage and were classed as “red-phase” due to some foliage retention. As woody branchwood fuels may become available for fire consumption due to MPB mortality in cases of extreme fire weather (Talucci and Krawchuck 2019), a secondary calculation of canopy fuel load with all diameters of branch wood was performed including the foliage component for live trees and branch biomass for MPB-killed trees.

Fig. 3. Forest mensuration sampling schematic adapted from the Next Generation Canadian Forest Fire Danger Rating System by Boucher et al. (2023) used as sampling methodology for data collection at 18 sites within Jasper National Park. Dashed blue lines indicate downed woody debris (DWD) transects (40 m each) over plot centre with diameter thresholds over segments. The purple circle indicates the overstory tree plot in which every mature and small tree (DBH ≥ 3 cm) was measured. The black circle indicates understory plot in which seedlings, saplings, and shrubs were measured. Small black squares indicate surface and ground fuel 1 m² microplots for surface coverage and organic layer depth.



Stand means of shrub and seedling fuel loadings were calculated using Jasper-specific equations from Delisle (1986). Downed woody debris loading was calculated using equations from McRae et al. (1979) with constants from Delisle and Woodard (1988). Downed woody debris is categorized into coarse woody debris (>7 cm diameter), medium woody debris (3 – 7 cm diameter), and fine woody debris (<3 cm diameter). Mean surface fuel load of moss and herbaceous material was calculated using % cover within plots and equations developed using data from the Kananaskis region of Alberta (approximately 500 km south of Jasper) as a regional proxy from Bessie and Johnson (1995). Litter and duff layer samples were dried and weighed for ground fuel loading for all plots.

Fuel strata gap was also calculated in each stand. Fuel strata gap is an adjustment of the canopy base height to include the effect of ladder fuels between the surface and the aerial fuels at a stand level introduced by Cruz et al. (2004). It has been recently adapted in a revised crown fire initiation modelling framework and is calculated by subtracting the centre of the subcanopy from the mean stand overstory canopy base height (Perrakis and Thompson 2025). Here, the centre of the mature spruce and fir and other small trees were subtracted from the overstory pine canopy base height. Stand age and outbreak severity extent was evaluated from a vegetation resource inventory vector GIS layer provided by Jasper National Park (British Columbia Forest Analysis and Inventory Branch 2024).

Statistical analysis

To better understand how the vertical distribution of fuels varied among MPB outbreak severity classes, we carried out a series of one-way nonparametric (Kruskal–Wallis) hypoth-

esis tests. These tests were applied to evaluate the effects of severity class on fuels attributes (i.e., ground, surface, understory, downed wood, aerial) and fuel structural components (i.e., height, canopy base height, fuel bed depth). The Kruskal–Wallis test (McDonald 2014; KW) is a nonparametric test appropriate for samples with non-normal distributions and unequal variances. Following identification of a significant KW model, the *post-hoc* Dunn's test was applied to evaluate which severity ratings were statistically different from one another for each fuel characteristic. All KW and Dunn's tests were performed in R (R Core Team 2023, version 4.3.1).

Results

Effects of gray-phase MPB on forest stand structure

The majority of measured MPB-affected stands were located along the Athabasca Valley south of Jasper townsite. Severity was not limited to a specific geographic area, but generally decreased in severity from north to south along the valley (Fig. 1B). The isolation of majority lodgepole stands and grouping the severity codes to the severity classes resulted in 23 970 ha classed as Light, 23,079 ha classed as Moderate, and 18 990 ha classed as Severe outbreak (Fig. 1B). Overall, 66 040 ha were found to have some level of MPB mortality across the park's extent (British Columbia Forest Analysis and Inventory Branch 2024).

Overstory structures and canopy fuels

Light MPB stands had a higher mean age of 156 years (standard error, SE = 25 years) than Moderate and Severe stands

Table 1. Mean ($\pm$ standard error (SE)) of stand characteristics determined from plot measurements and scaled per ha within forest stands with varying severity of MPB outbreak in Jasper National Park, AB. Means and severity rating highlighted in bold.

Outbreak severity	Total trees (stems ha ⁻¹)	Live basal area (m ² ha ⁻¹)	Dead basal area (m ² ha ⁻¹)	Quadratic mean diameter (cm)	Mountain pine beetle mortality (%)
Light (n = 5)	2275 (541)	23.1 (5.9)	4.1 (2.2)	16.7 (0.7)	18.0 (6.3)
Moderate (n = 8)	1550 (166)	13.8 (1.9)	8.6 (2.1)	16.8 (0.6)	54.6 (2.6)
Severe (n = 5)	810 (207)	7.9 (1.8)	8.8 (3.2)	18.5 (1.5)	76.0 (4.0)

Table 2. Species composition, stem density ($\pm$ SE) and proportion of living to dead pine across MPB outbreak severity for mature trees DBH $\geq$ 9 cm and small subcanopy trees (DBH 3–9 cm).

Outbreak severity	Total trees (stems ha ⁻¹)	Species proportions ^a	Pl trees (stems ha ⁻¹)	Percent live Pl	Percent dead Pl
Mature trees					
Light (n = 5)	1420 (424)	58 Pl; 40 Sx; 2 Bl	905 (330)	71	29
Moderate (n = 8)	1028 (127)	87 Pl; 3 Sx; 5 At; 4 Bl; 1 Dfi	919 (150)	36	64
Severe (n = 5)	660 (177)	96 Pl; 1 Sx; 3 Bl	635 (170)	18	82
Small trees					
Light (n = 5)	855 (146)	70 Sx; 17 Pl; 13 Bl	105 (29)	47	53
Moderate (n = 8)	521 (130)	49 Pl; 37 Bl; 14 Sx	219 (52)	15	85
Severe (n = 5)	150 (58)	90 Pl; 5 Bl; 4 Sx; 1 Dfi	135 (55)	13	87

^aSpecies abbreviation: Pl—Lodgpole Pine; Sx—Interior Spruce; Bl—Subalpine Fir; At—Trembling Aspen; Dfi—Interior Douglas Fir

which had a mean age of 105 (10), and 94 (14) years, respectively (British Columbia Forest Analysis and Inventory Branch 2024). Light stands had a higher overall mature (DBH > 9 cm) and small tree (3–9 cm DBH) density compared to Severe MPB stands. Light stands had a greater proportion of spruce intermixed with pine, and Moderate stands contained more subalpine fir and aspen than the other severity classes (Tables 1 and 2). Quadratic mean diameter was comparable in both Light and Moderate stands but slightly larger in the Severe stands (Table 1). Moderate and Severe MPB stands both had significantly lower live basal area relative to Light stands (KW $H = 6.61$, Dunn’s $z = -2.55$, $p < 0.05$), and twice the dead basal area as Light stands (Table 1), all of which indicate the effects of MPB severity on overstory structure and canopy fuels.

Tree height was not significantly different between MPB severity classes for overstory pine or subcanopy trees. Pine canopy base height was significantly higher in Light MPB stands (mean = 8.91 m; SE = 1.8 m) than Severe stands (mean = 3.8; SE = 0.64 m) (KW $H = 7.14$, Dunn’s $z = -2.49$, $p < 0.05$). Subcanopy (mature spruce, fir, and small trees) canopy base height did not differ between severity ratings. While Severe stands had overstory fuels that extended nearest to the ground, saplings and small trees in these stands were sparse.

Severe MPB stands had the lowest estimated canopy bulk density (mean = 0.02 kg m⁻³; SE = 0.01 kg m⁻³), which is a

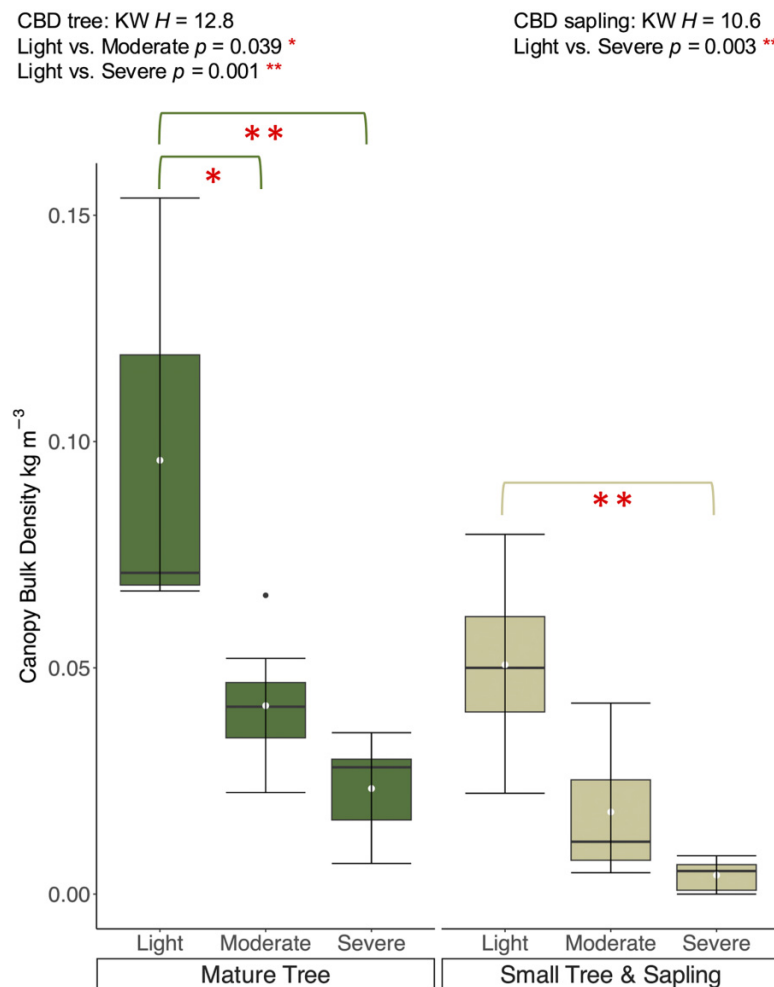
76% reduction compared to Light MPB stands (mean = 0.10 kg m⁻³; SE = 0.02 kg m⁻³, Fig. 4). Moderate MPB mean canopy bulk density (mean = 0.04 kg m⁻³; SE = 0.01 kg m⁻³) was slightly less than 50% of Light MPB stands (Fig. 4). In terms of proportional change from each severity classes’ pre-MPB condition, Light stands had lost a mean of 17%, Moderate stands lost a mean of 35%, and Severe stands lost 49% of original canopy bulk density.

Understory structures and canopy fuels

The density of small trees in the understory also varied greatly between stands (Table 2; central tendency: KW $H = 8.66$, Dunn’s $z = -2.55$, $p < 0.05$). Severe MPB stands had 96% fewer live small tree stems per hectare than Light mortality stands (central tendency: KW $H = 12$, Dunn’s $z = -3.41$, $p < 0.001$). In Moderate and Severe MPB stands, > 85% of pine stems in this size class were killed. As most epidemic outbreaks are considered on the decline when preferred host trees (>20 cm DBH) are exhausted, this corresponds to a prolonged outbreak where beetles start to attack smaller trees (Dhar et al. 2015).

The fuel strata gap between saplings and mature trees was significantly smaller in higher MPB severity stands than Light stands, but only when dead fuels were considered components of the aerial strata (central tendency: KW $H = 7.61$, Dunn’s $z = -2.67$, $p < 0.05$). Fuel strata gap often defines

Fig. 4. Canopy bulk density (CBD; kg m^{-3}) of mature trees and subcanopy (small trees and saplings) retaining foliage across a gradient of MPB-induced severity. Pairwise significance is denoted by red asterisks for central tendencies (Kruskal–Wallis with Dunn’s post hoc test: * = $p < 0.05$, ** = $p < 0.01$). Mean denoted by white dot.



canopy base height as height to the live crown, where fine aerial fuels (foliage) are sufficiently dense enough to enable crown fire initiation. In cases of prolific mortality due to outbreak, however, the dry bark and small twigs of beetle-killed trees may be able to fill this role. The lowest fuel strata gap was in Severe stands (mean = 1.38 m; SE = 0.49 m), Moderate had a slightly larger gap (mean = 3.94 m; SE = 0.68 m) and Light had the largest (mean = 5.74 m; SE = 1.6 m).

The cohort of small trees and saplings in the Light MPB severity class understory were the tallest and most dense, and also had the greatest canopy bulk density among understory strata (mean = 0.05 kg m^{-3} ; SE = 0.01 kg m^{-3}). The small-stemmed pines also experienced high levels of mortality due to MPB (Table 2), resulting in small canopy bulk density values for Moderate stands (mean = 0.02 kg m^{-3} ; SE = 0.01 kg m^{-3}), and negligible values for Severe stands which had higher proportion of pines in the regenerating conifers cohort (mean = 0.004 kg m^{-3} ; SE = 0.002 kg m^{-3} , Fig. 4). From pre-MPB condition canopy fuel loading, Light MPB-affected stands lost 6%, Moderate stands lost 19%, and Severe stands lost 49%. Foliage retention in the Light MPB-affected subcanopy cohort is due to the higher density of

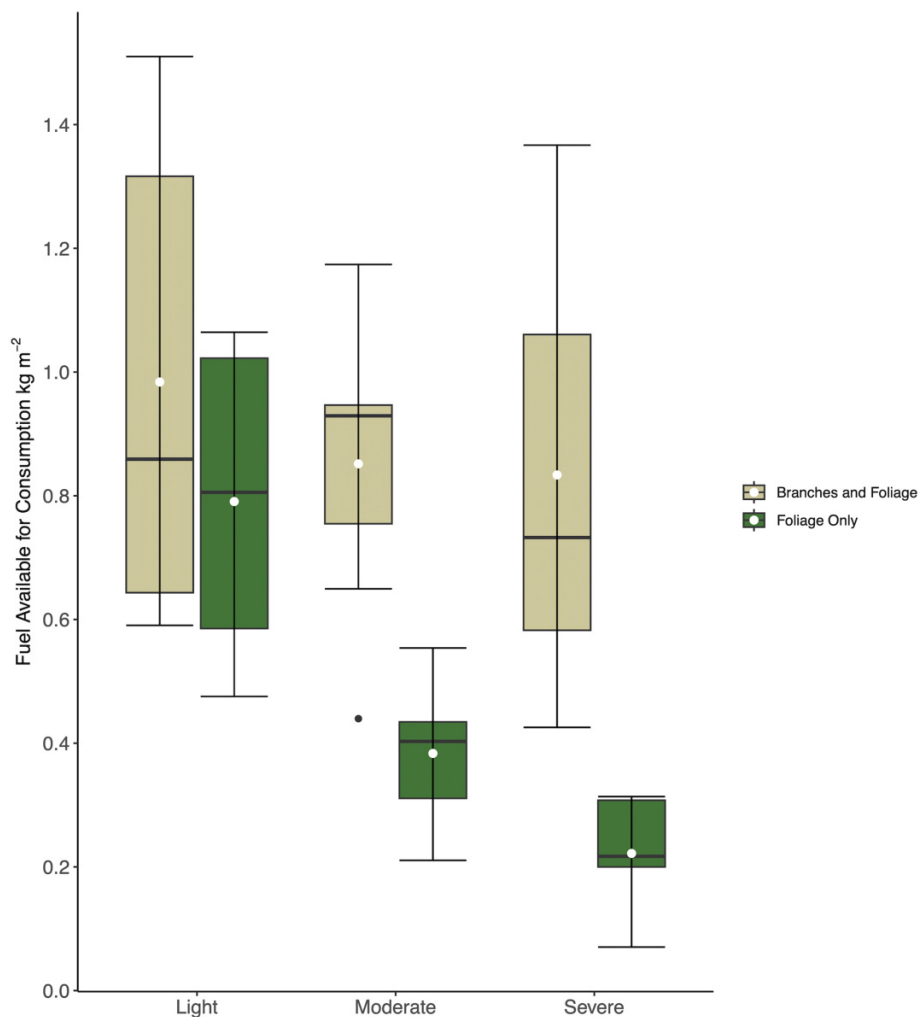
spruce in the understory (Table 2), and likely due to site conditions (e.g., higher moisture).

Increase in potential fuel availability

To show the potential change in biomass in the canopy available for consumption in extreme fire weather scenarios, we compared foliage-only and foliage-plus-branches scenarios using biomass estimates from allometry. Foliage-plus-branches includes the biomass of the branchwood of MPB-killed trees, as the change in fuel moisture may allow for these fuels to be available for consumption in dry conditions. The greatest change in estimated available fuel was seen in Severe stands, increasing from 0.22 kg m^{-2} to 0.83 kg m^{-2} (SE = 0.17 kg m^{-2}), a 3.8-fold increase due to branch biomass. Moderate stands' available fuel increased using this estimate just over two times from 0.38 to 0.85 kg m^{-2} (SE = 0.08 kg m^{-2}), a 122% increase (Fig. 5). Light stands experienced a 24% increase from 0.79 to 0.98 kg m^{-2} (SE = 0.18 kg m^{-2}) (Fig. 5).

The distribution of coarse and fine downed woody debris was not found to be significantly affected by MPB severity class. However, Moderate sites had the highest loading of coarse woody debris (mean = 3.21 kg m^{-2} ; SE = 0.49 kg m^{-2}).

Fig. 5. Modelled stand canopy fuel biomass expected to be available to burn in extreme fire weather scenarios based on foliage (green) and foliage with branch wood (tan) (for MPB-killed trees) across MPB outbreak based on allometric equations per plot. Mean denoted by white dot. Beetle-killed trees branches may become available to fire consumption given high fire weather indices (Talucci and Krawchuck 2019). Differences between treatments were not significant for branches + foliage at the $\alpha = 0.05$ level.



m^{-2}) approximately 1 kg m^{-2} more than both Light and Severe sites (Fig. 6), which may be of ecological significance. This is likely associated with the combination of higher average stem density than Severe and more mortality than Light severity stands. As Moderate and Severe stands have comparable dead standing basal area (Table 1), this ratio will likely remain constant as dead pines fall to the forest floor, in areas not impacted by fire.

Effects of canopy opening on surface community composition and loading

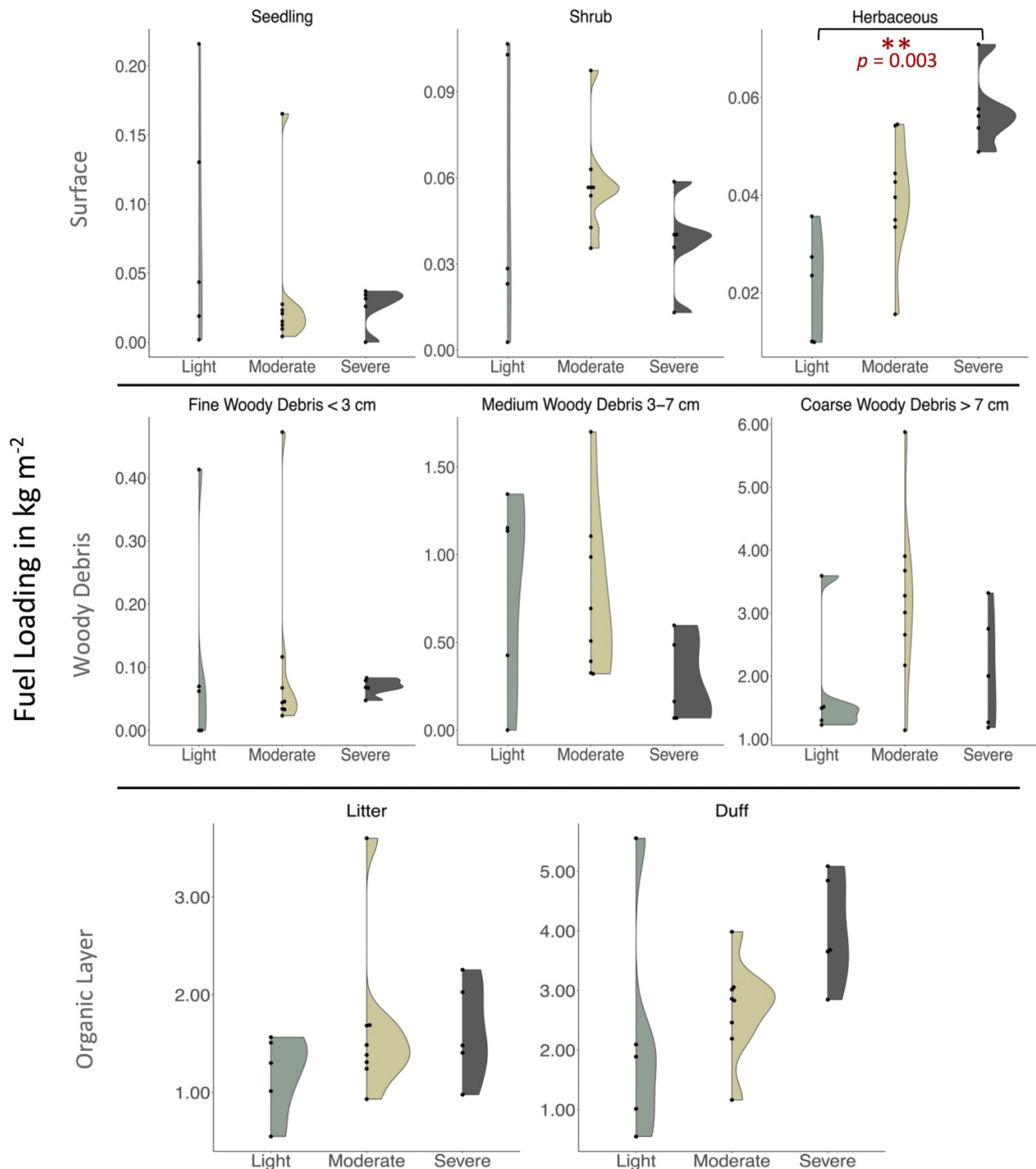
Coniferous regeneration varied in species composition and productivity (Table 3). Severe MPB stands had greater % coverage of herbaceous and grass vegetation than Light and Moderate stands (Table 4). Light MPB stands had the most conifer seedlings per hectare (Table 3). Moderate and Severe stands had lower mean depth organic material (litter and duff) compared with Light stands (Table 4), but this may be a func-

tion of plot location. Light severity stands had the highest seedling fuel load (total biomass, foliage and woody material; mean = 0.08 kg m^{-2} ; SE = 0.03 kg m^{-2}), over two and three times greater than Moderate and Severe MPB stands, respectively (Fig. 6), though this was not significant. Differences in shrub loading and coverage were not significant (KW $H = 1.91$, $p = 0.38$). Grass and herbaceous fuel loading were greater in Severe than Light MPB stands (KW $H = 11.0$, Dunn's $z = 3.32$, $p < 0.005$) (mean = 0.06 kg m^{-2} and 0.02 kg m^{-2} , respectively). The weight of litter did not significantly vary between MPB severity classes. The dry weight of the duff layer, however, was approximately twice as much in Severe stands (mean = 4.0 kg m^{-2} ; SE = 0.41 kg m^{-2}) as Moderate and Light MPB stands (Fig. 6).

Discussion

This study examined the diversity in fuel structures in gray post-MPB phase (4–10 years since attack) across three

Fig. 6. Fuel loading (kg m^{-2}) histogram distribution of understory fuels, downed woody debris (fine < 3 cm, medium 3–7 cm, and coarse > 7 cm diameter) and organic layer (litter and duff—fermentation and humic layers) by MPB severity rating. Significant differences between severity levels are denoted in red (Kruskal Wallis: $** = p < 0.01$).



MPB outbreak severity classes. Overall, the major contributors to variation in stand and fuel characteristics as outbreak severity increased were in the affected understory small tree (advanced regeneration) cohort, dead standing basal area, canopy fuels, and herbaceous vegetation coverage (Table 1;

Table 2; Fig. 7). Severe stands within this study lacked recruitment of conifer seedlings, and had lower shrub and moss coverage, and higher herbaceous vegetation fuel loadings (Table 2; Fig. 6). Moderate stands had the greatest amount of coarse woody debris (Fig. 6) but exhibited the same amount

Table 3. Stem density and species composition of advanced and new coniferous regeneration across MPB outbreak severity for saplings DBH > 3 cm and seedlings (<1.37 m height).

MPB outbreak severity	Total trees (stems ha ⁻¹)	Species proportions ^a
Saplings		
Light (n = 5)	485 (163)	91 Sx; 5 Pl; 4 Bl
Moderate (n = 8)	200 (146)	50 Fir; 33 Pl; 17 Bl
Severe (n = 5)	80 (49)	50 Sx; 50 Bl
Seedlings		
Light (n = 5)	3240 (1430)	82 Sx; 16 Pl; 2 Bl
Moderate (n = 8)	1325 (649)	74 Pl; 16 Bl; 10 Sx
Severe (n = 5)	1000 (283)	68 Pl; 25 Sx; 7 Bl

^aSpecies abbreviation: Pl—Lodgpole Pine; Sx—Interior spruce; Bl—subalpine Fir; At—Trembling Aspen; Dfi—Interior Douglas Fir

of dead standing basal area as Severe stands (Table 1). Moderate stands therefore may present high levels of potential fire hazard in terms of impaired firefighter maneuverability (Moriarty et al. 2019), and increased surface fire intensity potential when transitory dead standing fuel falls (Schoennagel et al. 2012). Severe stands may exhibit higher potential for fire ignitions due to lack of canopy fuels increasing solar radiation and in stand winds to the forest floor (Jenkins et al. 2008). Loss of canopy fuels in both overstory and understory fuel strata in Severe stands represents a potential lower crown fire hazard (Hicke et al. 2012). Nevertheless, dead standing snags and increased fuel availability in these stands represent alternative fuel hazards that likely contributed to fire behaviour in the Jasper fires of July 2024.

Mortality influences vertical canopy fuel distribution

Relative to initial stand conditions, overstory canopy bulk density (foliage) decreased 49% in Severe stands, compared to 35% in Moderate and 17% in Light severity stands (Fig. 4). Canopy bulk density is one of the most important factors in crown fire spread (Van Wagner 1977). Continuity of fine aerial fuels is considered necessary to sustain active crown fire. Sparse canopy fuels (low canopy bulk density and canopy cover) can increase incident solar radiation to the understory and forest floor (Whitehead et al. 2008) and increased spacing between trees can alter the roughness length, or wind obstruction level, thereby increasing atmospheric turbulence and stomatal conductance and evaporative losses from lower canopy and understory (Chasmer et al. 2008). Combined, these drying influences can decrease fuel moisture, increasing ignition potential and surface fire behaviour (Wotton and Beverly 2007).

The effect of MPB outbreak severity is especially pronounced in the small tree cohort with the relative loss of canopy fuels. Pre-outbreak species composition is playing an

apparent role in differences between the mixed-conifer Light and more pure pine Moderate and Severe stands (Table 2), but mortality is seen in the small pines in all severity classes despite being lower than the preferred MPB stem diameter (> 20 cm). Severe stands lost an average of 66% of canopy bulk density compared to pre-MPB condition, as opposed to only a 6% relative loss in Light stands. This finding aligns with the notion that there will be a surviving secondary structure after Light and Moderate MPB outbreak severity which alters even-aged pine stands into multi-level cohort stands (Axelson et al. 2009; Hansen et al. 2015), but the ecosystem will be more severely altered in Severe stands. The loss of the understory small tree cohort may contribute to a significant time lag before regeneration will become sufficient to contribute to ladder fuels and increase crown fire potential in Severe stands, as seedling densities were also low in these stands.

Despite loss of canopy fuels, extreme fire weather scenarios can make standing MPB-killed branch wood available for burning and fuel consumption in gray-phase MPB stands (Talucci and Krawchuck 2019). If this fuel is consumed in an intense surface fire or fire with canopy involvement, the amount of available fuel surpasses the amount of foliage fuels lost from the canopy (Fig. 5). Fuel consumption of these branch structures may occur due to smouldering fire behaviour, thus not contributing to intensity and spread in the flaming front, but still influences fire hazard from a suppression perspective, and alters post-fire structural legacies. Talucci et al. (2022) also found that in Interior British Columbia, fire severity was highest in mixed live and dead vegetation, suggesting the fuel profile found in Moderate stands here may represent higher fire severity risk.

Canopy openness shifts montane vegetation communities

The higher level of herbaceous coverage and loading found in Severe stands aligns with an increase in canopy openness; this could also explain the decrease in moisture- and shade-dependant moss species coverage. Pine regeneration postepidemic in warm climates and drier sites has been noted to increase with severity levels (Pelz et al. 2018). Despite this, there is limited seedling recruitment as severity increases (Table 2). Closer to this study area, however, poor regeneration of pine seedlings has been noted 1–3 years post MPB mortality (McIntosh and Macdonald 2013) in Robb, Alberta (76 km east of Jasper) as well as 6–9 years post MPB mortality in the boreal region near Grande Prairie, AB (400 km north of Jasper; Lieffers et al. 2023). Here, high severity stands exhibiting low recruitment may result in an ecological shift to other coniferous, broadleaf, or nonforest herbaceous and grass species dominated ecosystems (Whitman et al. 2019; Lieffers et al. 2023). In Jasper, such shifts could contribute to the historical mosaic of conifer, aspen and grassland stands (Tande 1979), promoting heterogeneity across the landscape.

Surface and canopy fuel implications for potential fire behaviour

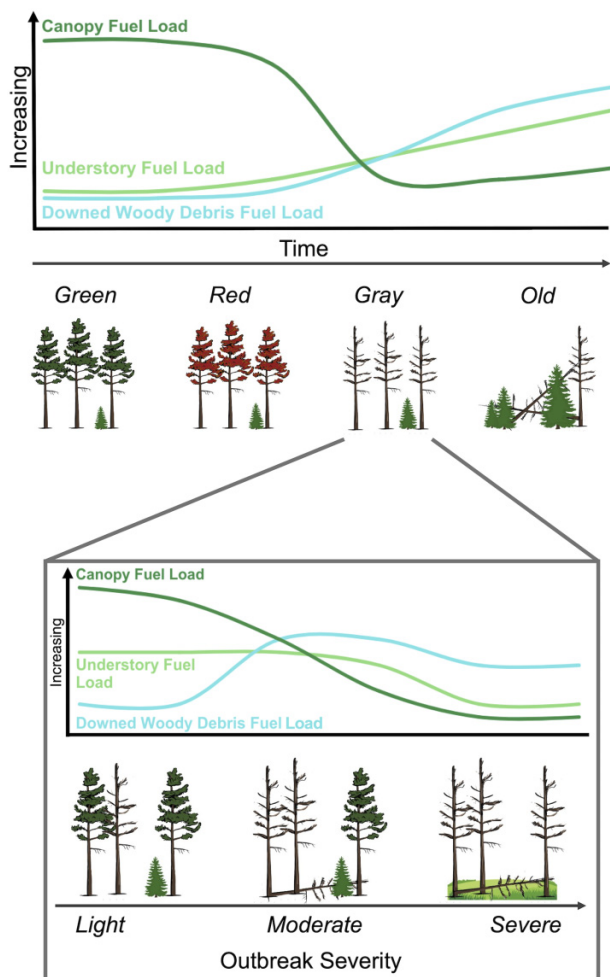
Though many surface fuel variables were not found to differ statistically among stands, the varying ranges may

Table 4. Mean ($\pm$ SE) understory characteristics of stands with varying severity of MPB outbreak in Jasper National Park, AB.

MPB outbreak severity	Shrub cover (%)	Herbaceous cover (%)	Moss cover (%)	Litter cover (%)	Organic layer depth (cm)
Light (n = 5)	40.6 (17.2)	22.8^a (3.4)	70.5 (16.8)	35.8 (14.1)	17.8^b (5.8)
Moderate (n = 8)	57.5 (4)	37.8^a (5.7)	75.4 (6.3)	12.2 (3)	6.4^a (1.1)
Severe (n = 5)	31 (9.8)	61^b (4.7)	39.9 (13)	8.8 (2.5)	6.4^a (1.3)

Note: Values with different letters (KW with Dunn's post-hoc test) are significantly different for that surface fuel component.

Fig. 7. Conceptual framework of canopy, understory, and downed woody debris fuel loading levels over traditional time since beetle phases, and along varying mortality levels within the gray-phase.



have implications for fire and fuel considerations. Herbaceous foliage found in Severe stands can retain more moisture than woody stemmed species but also exhibit more extremes through seasonal curing and less moisture storage in root systems (Loudermilk et al. 2022). The highest duff loading was found in Severe stands (Fig. 6), which can contribute to longer fire smouldering and residence time (Cowan et al. 2020).

Severity classes were not found to have a significant effect on loading of coarse woody debris (Fig. 6). Given the difference in standing dead basal area (Table 1), however, this study establishes an important baseline of downed woody debris *potential* for MPB-affected stands in the Canadian Rocky Mountains; when fires occur, heavy woody fuel loading increases the potential for intense fire behaviour and high-risk and inaccessible scenarios for suppression operations (Page et al. 2013). Standing snags also burn unpredictably, as dry, weakened wood can fall in unexpected ways and create large fire brands from fissured branches (Moriarty et al. 2019). As time since outbreak increases, a large contributor to fuel profile change is falling snags. A magnitude of change in coarse woody debris in these stands was not yet apparent (Fig. 6), but as these standing snags are subject to snowfall, windthrow, and wildlife activity, the transition of many snags to the forest floor will be an important consideration for fire-fighting operations as well as the public at large (Audley et al. 2021; Fig. 7). The timing of fall and decomposition can vary depending on regional climatic variables such as wind fields, temperature, and precipitation patterns. In the western U.S., a study predicted 75% of snags to fall by 22 years post-outbreak (Audley et al. 2021). In Montana, areas with little topographical relief and low stem densities were found to have shorter snag persistence and found 40% of beetle-killed trees had fallen seven years after the peak of infestation (Latif et al. 2023). With most of this studies' stands approximately four years post-peak infestation, and 40.7% and 53% of all standing stems as snags in moderate and severely affected stands, respectively (Table 1), the coarse woody loading can be expected to increase substantially in the next 5–10 years.

Though Light stands experienced a small amount of overstory mortality, they did not experience nearly as much relative midstory mortality as Moderate and Severe stands (Table 1). Fire behaviour in gray-phase MPB has been observed by one of the authors (DDBP) to burn as "mixed fire-behaviour", where abundant torching occurs into remnant aerial fine fuels, and some involvement of standing woody fuels is apparent, but the fire spread mechanism resembles intense surface fire across most of the landscape. However, extreme weather or wind-dominated scenarios, coupled with the aridity found in dead standing fuels, can create fire behaviour and spread that is difficult to predict based on past research that used modelling systems built on results from more moderate conditions. While these measurements represent the condition of the pre-fire environment, much of the fuels in the landscape we studied was consumed in a few

hours in the intense Jasper wildfire in late July 2024. We have left our fuels analysis unbiased by this event, but it is a reminder of the importance surrounding the intricacies of fire behaviour prediction.

Limitations and future research

In this study, two forest plots in the Severe rating class exhibited lower initial stand density than all other plots, and therefore may influence the perceived reduction of canopy fuels. To take this into consideration, relative loss from the initial stand condition was calculated and presented alongside the between-rating comparisons. Another potential limitation was number of plots measured. Although 18 plots may not be optimal for statistical strength, high-quality data collection was constrained to this valley to limit the inherent variability found within forest complexes, which still proved challenging. Utilization of the vegetation resource index for severity rating coverages showed each severity class is relatively even across the outbreak extent in the park, with slightly less area covered by Severe stands (Fig. 1B). Coincident airborne lidar and other remote sensing data has been collected in this region, and using these field data as ground validation will allow for scaling these findings to the landscape level. The Jasper fire provides an opportunity to examine potential relationships between fuel characteristics and the extreme fire behaviour exhibited. Future research examining insect disturbed fuels can use these MPB severity thresholds to link disturbance severity to potential fuel loading in comparable pine forest complexes.

Conclusions

Though much research has been done to examine the differences in forest structure and fuels as time since MPB attack increases, this study highlights the variability observed within the same temporal stage (i.e., “gray-phase”) separated into three different outbreak severity classes based on percentage mortality within a stand. The cumulative MPB outbreak in western Canada has led to the need for updated classification systems for severity and more regional documentation of gray-phase MPB effect on fuels. The mortality thresholds reported here (Light < 35%, Moderate 36%–65%, and Severe > 65%) showed variation in fuel structure and loading along vertical strata, especially in the small tree or subcanopy cohort. The differences found in canopy fuels, small tree cohort loss, surface composition, and organic layer loading drive these differences in severity class fuels. An increase in herbaceous vegetation coverage and lack of conifer seedling recruitment suggest that reversion to open forest, mixed conifer, or grassland communities may occur where MPB mortality has been most severe, even in the absence of fire. Beyond the potential for increased wildfire intensity, care is needed in Moderate and Severe MPB-affected stands going forward as the prevalence of dead fuel presents increased hazards for the public using these forests for recreation as well as firefighting and management personnel in operational activities.

Acknowledgements

Many thanks to field team members Dr. Zhouxin Xi, Emily Jones, Saeid Parsian, Dr. Linda Flade, Viviana Lartiga, Italo Rodrigues, Jessica van Gaalen, and Jeanne Franco! We would also like to thank Dr. Chris Hopkinson for the use of vehicle and survey equipment.

Article information

History dates

Received: 6 December 2024

Accepted: 2 March 2025

Accepted manuscript online: 17 March 2025

Version of record online: 2 April 2025

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Data availability

Data generated or analysed during this study are available from Skretting and Chasmer.

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Patrick M.A. James served as Associate Editor at the time of manuscript review and acceptance; peer review and editorial decisions regarding this manuscript were handled by another editorial board member.

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Competing interests

The authors declare there are no known competing interests that could influence this paper.

Funding information

This project was supported by fRI Research Federal Provincial MPB Research Partnership (grant number 247.15) to Chasmer. Funding for fieldwork was supplemented by NSERC Canada Wildfire (grant number SPG-N NETGP 548629-19; Chasmer), and Parks Canada—Jasper National Park and Alberta Innovates (Dr. Chris Hopkinson; Chasmer).

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